

ST BENEDICT'S COLLEGE



SUBJECT	AP MATHEMATICS	DATE	03 November 2015
GRADE	11	MARKS	100
EXAMINER	Mr Benecke	MODERATOR	Mrs. Povall and Mr. Gill
NAME		DURATION	1 hour
CLASS			

QUESTION NUMBER	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Conditional Probability	15	
2	Probability Theory	10	
3	Combinations and Permutations	15	
4	Probability Distributions	25	
5	Normal Distributions	25	
6	The Least Squares Regression Line	10	
	Total	100	

1. This paper consists of 9 pages
2. Write your answers on this paper.
3. Round off all answers to **2 decimal places** unless otherwise stated
4. Calculators may be used.
5. Extra Space has been provided on page 6 for rough work.

QUESTION 1

15 MARKS

The probability that a St Benedict's boy plays cricket is 0.45 while the probability that he plays rugby is 0.65. If the probability that he plays neither sport is 0.05 then determine:

a) The probability that he plays both Cricket and Rugby? (5)

b) The probability that he plays only Cricket? (2)

c) The probability that he plays Rugby or Cricket? (2)

d) $P(\text{Cricket} \mid \text{Rugby})$ (2)

e) Are the events "playing Cricket" and "playing Rugby" independent? Justify your answer. (4)

QUESTION 2**10 MARKS**

a) Let A and B be Complementary Events.

i) Determine $P(A \cap B)$ (2)

ii) Determine $P(A)$ if $P(B) = \frac{3}{5}$ (2)

b) Let C and D be Exhaustive events where $P(C|D) = \frac{2}{5}$ and $P(D) = \frac{3}{10}$.

i) Determine $P(C \cap D)$. (3)

ii) Determine $P(C)$. (3)

QUESTION 3

15 MARKS

Consider the word SIYAVULA (Repeated letters are treated as identical):

- a) How many different eight-letter word arrangements can be made? (3)

- b) What is the probability that the word will start and end with the same letter? (3)

- c) What is the probability that the word will start with S end with Y? (3)

- d) What is the probability that the A's will be next to each other? (3)

- e) If I select only 4 letters, what is the probability that all four letters selected will be vowels?
(Hint: Y is not a vowel) (3)

QUESTION 4

25 MARKS

a) A six-sided fair die is thrown 10 times, what is the probability that:

i) The number “6” appears twice? (4)

ii) The number “6” appears at least twice? (6)

b) A bag contains 5 red balls, 3 black balls, 7 white balls, and 5 yellow balls. If 4 balls are taken simultaneously what is the probability that:

i) All four balls will be red? (4)

ii) One of each colour is taken. (5)

- c) A car factory builds 100 cars every day, 15 of the cars built today had electrical problems. If a sample of 10 cars were to be taken from today's stock, what is the probability that 2 cars would be found to be defective? (6)

(Extra Space for Rough Work)

QUESTION 5**25 MARKS**

- a) The heights of all the Basketball players in the NBA (National Basketball Association) were measured and it was found that their heights were normally distributed with a mean of 200cm and a standard deviation of 8cm. What is the probability that a player chosen of random will be:

i) Taller than 210cm? (7)

ii) Between 195cm and 199cm tall? (8)

- b) The heights of all the Basketball players in the ABL (African Basketball League) were measured and it was found that their heights were normally distributed with a mean of μ cm and a standard deviation of σ cm. It was also found that 10% of the players were taller than 205cm and 9% of the players were shorter than 195cm. Calculate the values of μ and σ . (10)

QUESTION 6**10 MARKS**

To test the effect that sleep has on exams, a sample of 10 students was selected. They were asked to write a mathematics exam before sleeping and then after having slept. The exam was scored out of 100.

The results were as follows:

Student	1	2	3	4	5	6	7	8	9	10
Initial Score (I)	75	72	72	68	81	84	70	78	86	82
Final Score (F)	90	83	68	70	92	95	65	80	92	79

- a) Determine the correlation coefficient. (2)
- b) What does this tell us about the nature and the strength of the linear relationship between I and F. (2)
- c) Determine the equation of the least squares regression line of F on I. (2)
- d) Two further students Mark and Luke are tested and their initial scores are 85 and 64 respectively.
- 1) Predict their final scores. (2)
- 2) Which of these predictions would be more reliable? Why? (2)